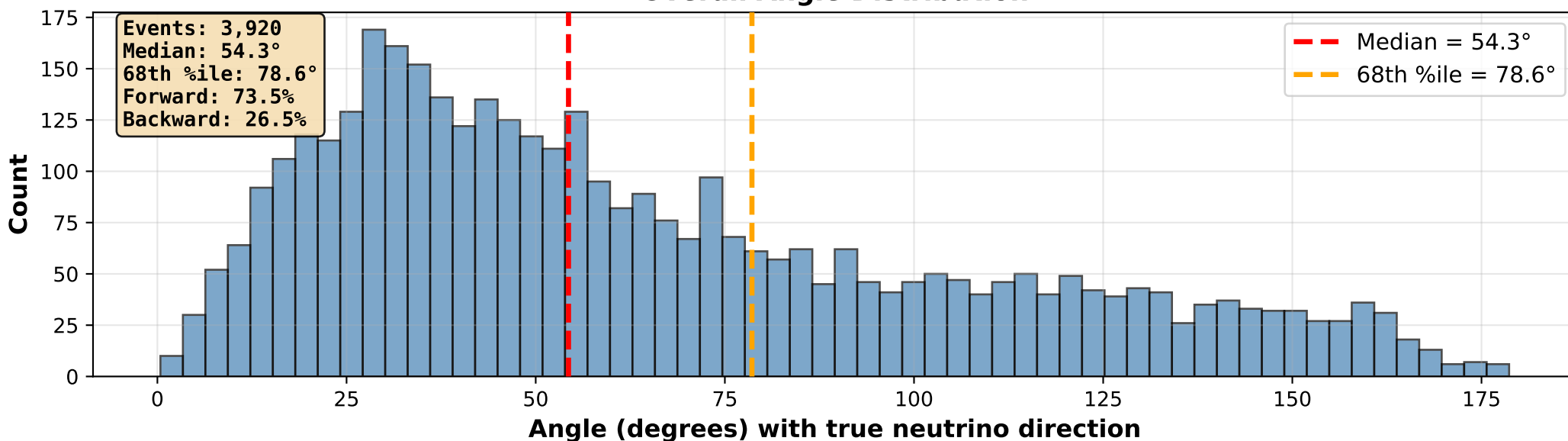


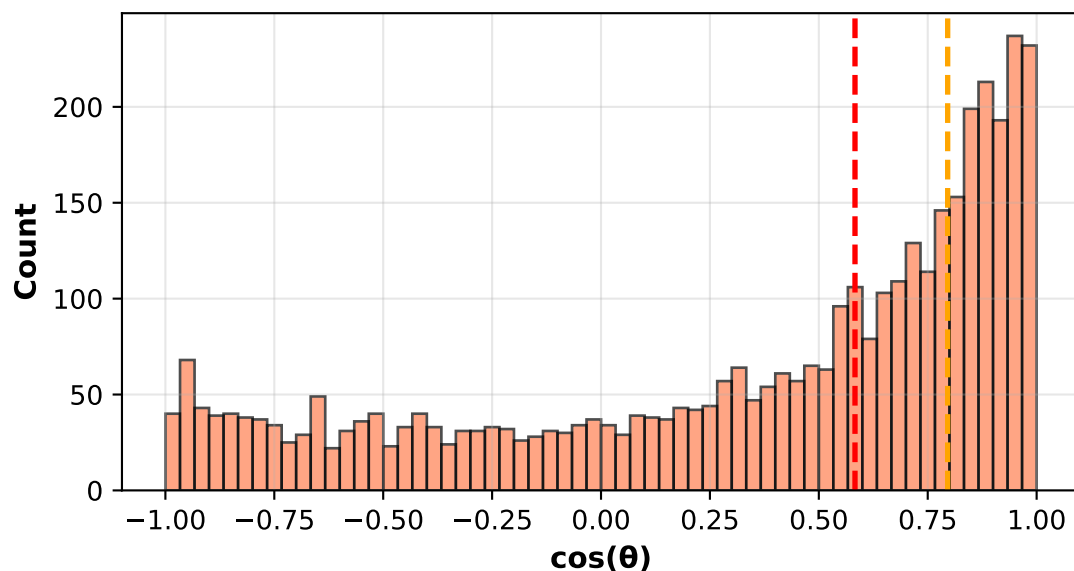
ED Model + MCMC Pointing Performance

Overall Results (98 categories, 3,920 events)

Overall Angle Distribution



Cosine Distribution



SUMMARY STATISTICS

Total Events: 3,920
Successful Categories: 98

Angle Distribution:
Median: 54.3°
Mean: 65.4°
68th percentile: 78.6°
Std deviation: 42.1°

Pointing Direction:
Forward ($\cos > 0$): 2,883 (73.5%)
Backward ($\cos < 0$): 1,037 (26.5%)

COMPARISON TO BASELINE

Best possible (true e^- dirs):
Angle 68% = 4.4°

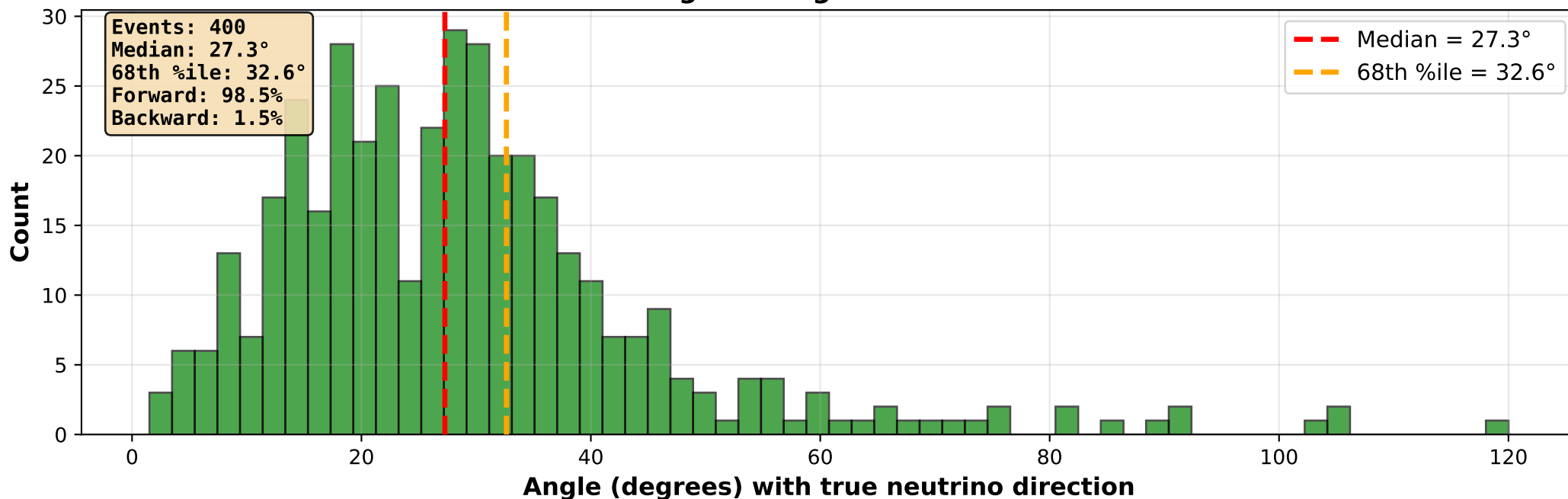
Current (ED model):
Angle 68% = 78.6°

ED Model + MCMC Pointing Performance

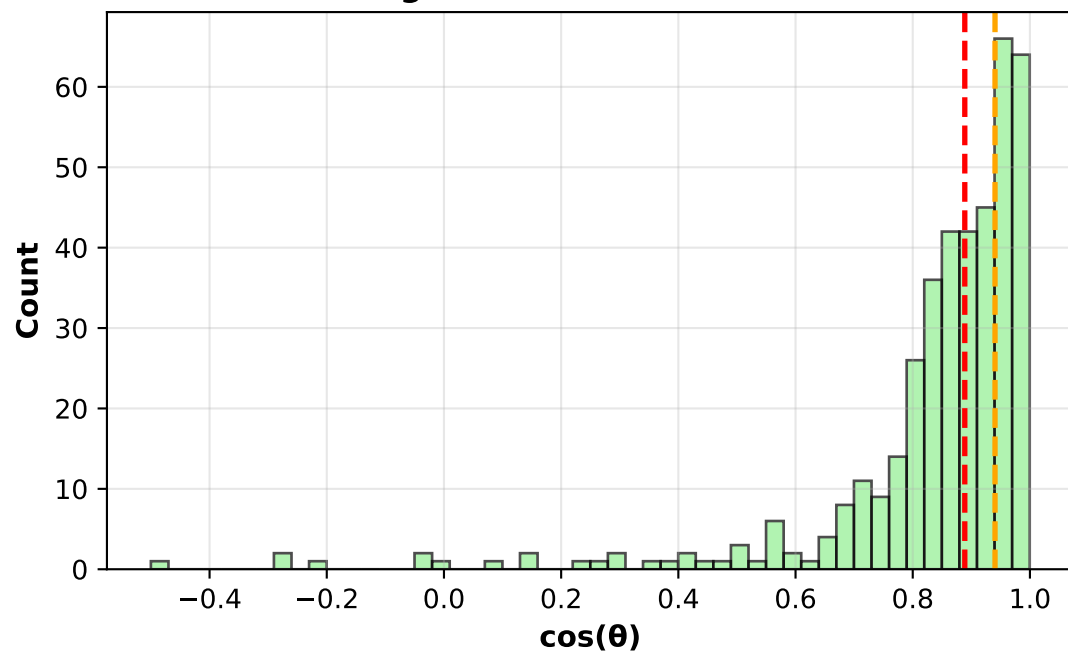
Z-aligned (within $\pm 30^\circ$ of axis)

10 categories, 400 events

Z-aligned: Angle Distribution



Z-aligned: Cosine Distribution

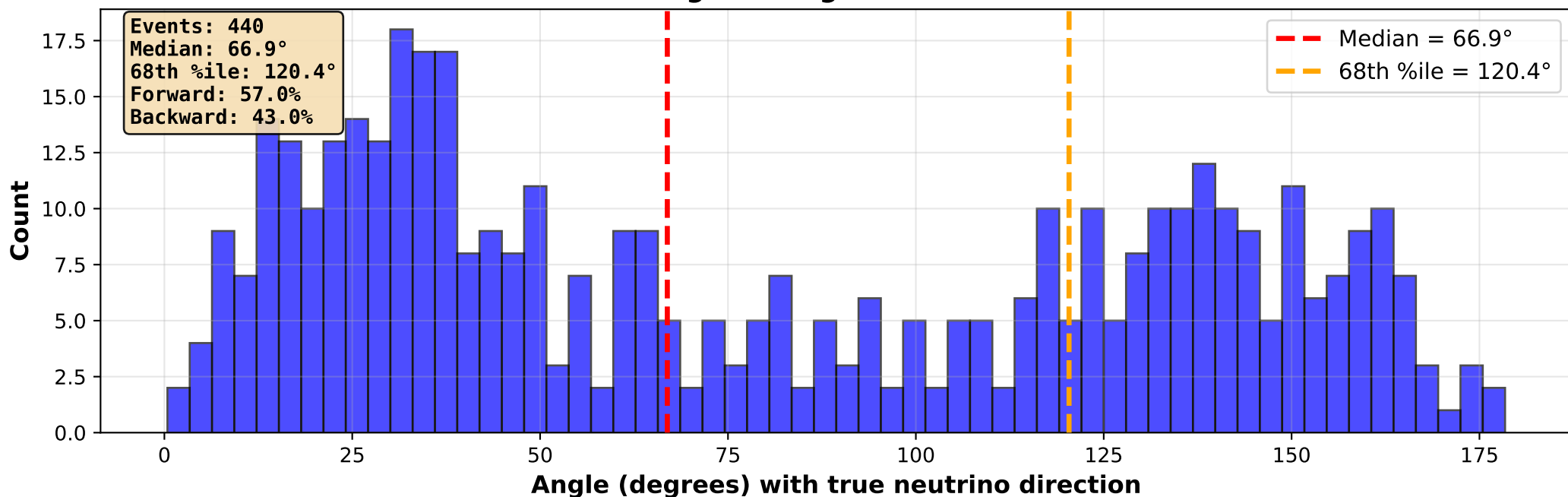


ED Model + MCMC Pointing Performance

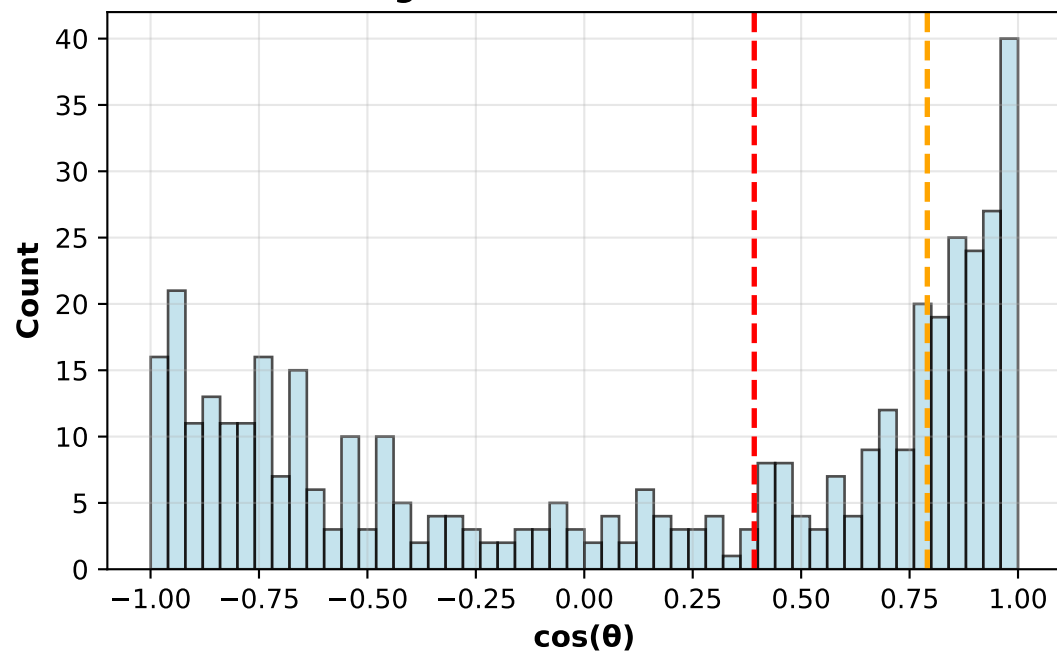
X-aligned (within $\pm 30^\circ$ of axis)

11 categories, 440 events

X-aligned: Angle Distribution



X-aligned: Cosine Distribution

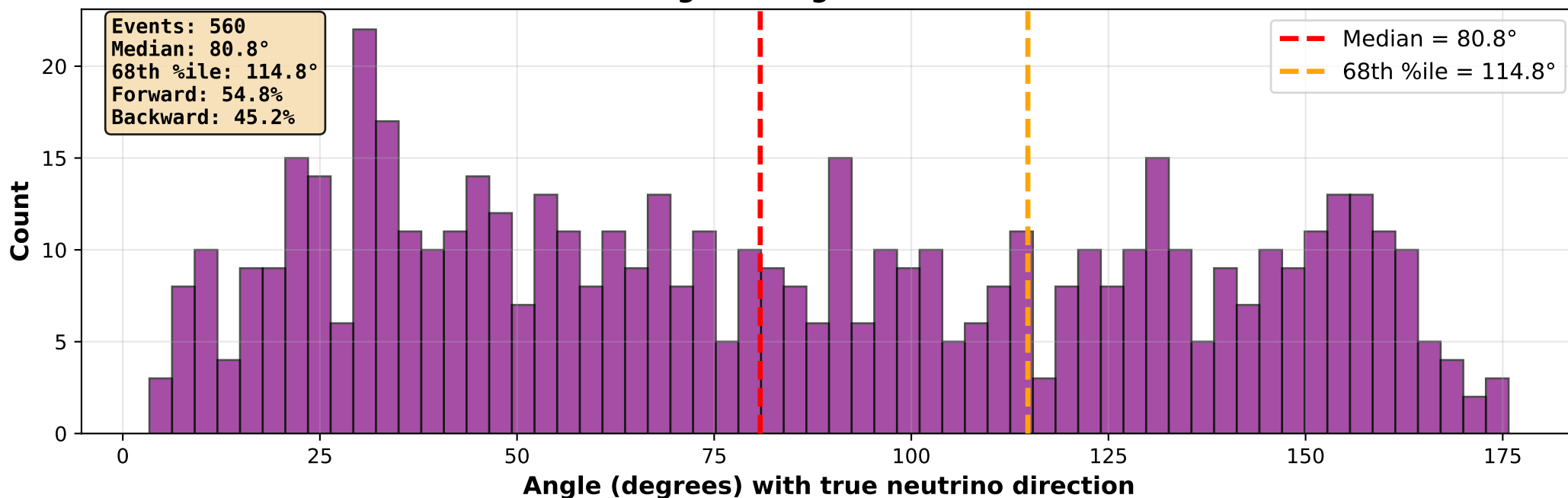


ED Model + MCMC Pointing Performance

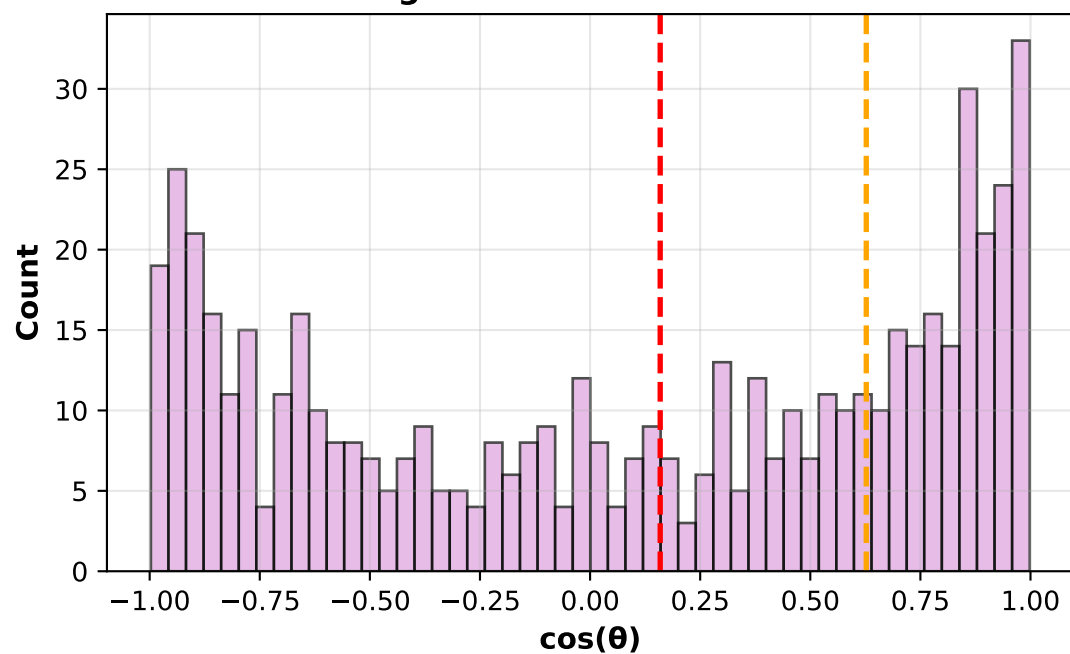
Y-aligned (within $\pm 30^\circ$ of axis)

14 categories, 560 events

Y-aligned: Angle Distribution



Y-aligned: Cosine Distribution

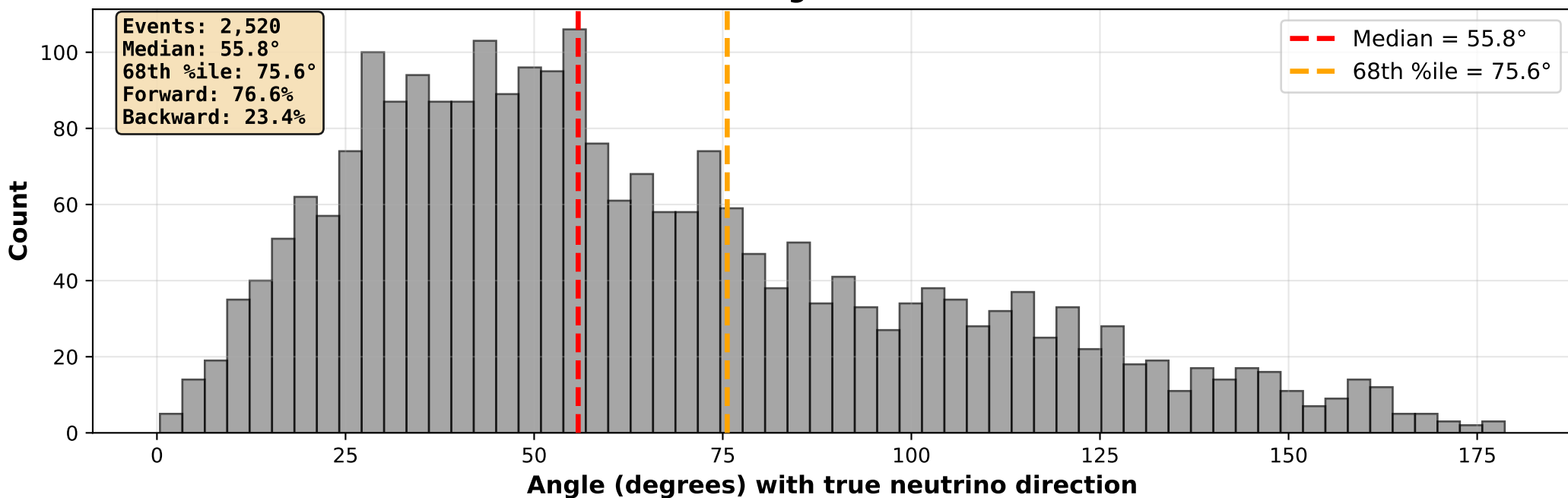


ED Model + MCMC Pointing Performance

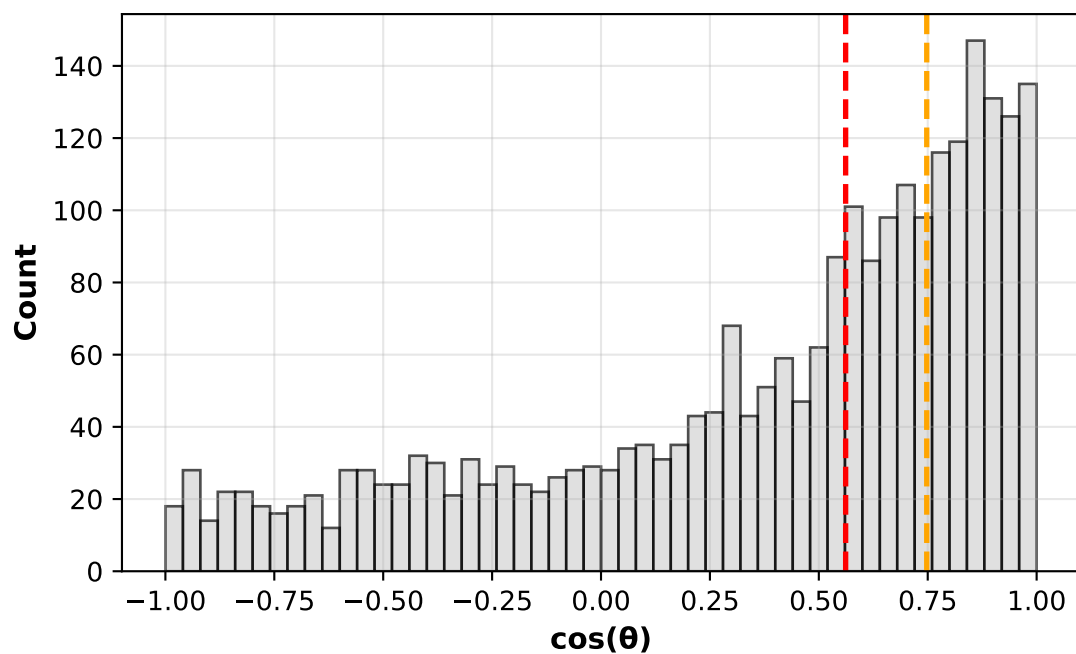
Other (not aligned with any axis)

63 categories, 2,520 events

Other: Angle Distribution

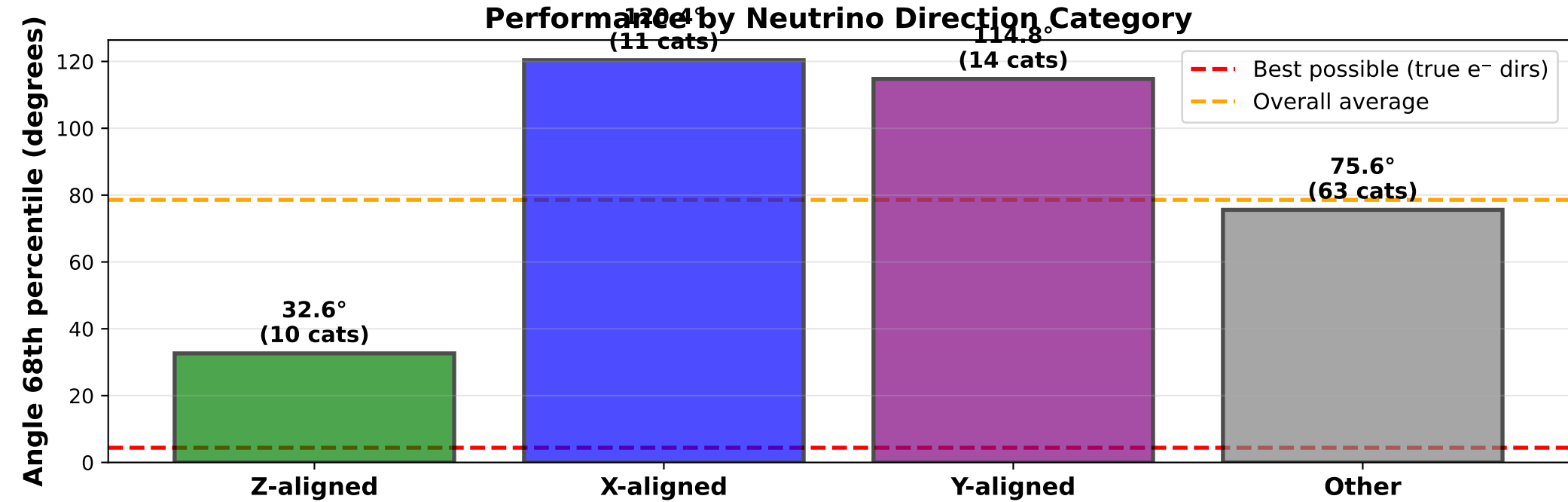


Other: Cosine Distribution



ED Model + MCMC Pointing Performance

Performance Comparison by Neutrino Direction



Performance Summary by Direction					
Category	N_cats	N_events	Angle 68%	Median°	Forward%
Z-aligned	10	400	32.6°	27.3°	98.5%
X-aligned	11	440	120.4°	66.9°	57.0%
Y-aligned	14	560	114.8°	80.8°	54.8%
Other	63	2,520	75.6°	55.8°	76.6%
Overall	98	3,920	78.6°	54.3°	73.5%

Key Findings	
• Performance strongly depends on neutrino direction:	
- Z-aligned: 32.6°	
- X-aligned: 120.4°	
- Y-aligned: 114.8°	
- Other: 75.6°	
• Best possible (true e^- directions): 4.4°	
• Current gap: Δ angle = 74.2°	